

ClaimLens

The claims AI that knows when it doesn't know

Adversarial debate and calibrated confidence for insurance claims triage

HIGH

MEDIUM

LOW

A \$308 billion problem — and a capacity problem

US insurance fraud losses, 2025 estimate

\$308B

Estimated annual US insurance fraud losses across all lines

75%

of claims flagged as suspicious are never fully investigated — the bottleneck is investigator capacity, not detection

2,137%

rise in deepfake fraud attempts since 2023; AI-enabled document fraud up 3,000%

Insurers lose \$8.50 for every \$1 spent fighting fraud. Fraud has gone AI-powered — the defence has not.

The obvious build fails in a specific way

We built a single-pass LLM decider as our baseline. It is confident and wrong.

Wrongly APPROVE a fraudulent claim

Direct financial loss. Recoverable, measurable, and priced into the business.

Wrongly DENY a legitimate claim

Customer harm, churn, regulatory exposure and bad-faith liability. Not recoverable.

An AI that is 85% confident of both is unusable in a regulated industry.

Confidence has to mean something before automation is safe. That is the problem ClaimLens is built around.

How ClaimLens works

It declares how sure it is before anything is automated

1

Evidence

Deterministic case file. Policy timeline, claim history, document consistency, live weather lookup. Facts only, no conclusions.

2

Debate

A Prosecutor argues the claim is suspicious. A Defender answers each point.

3

Judge ×2

The same arguments judged twice, in opposite orders. A verdict that flips with order is not a verdict.

4

Accounting

Each point marked answered-with-fact, answered-by-assertion, or unanswered.

5

Gate

Confidence capped in code. Auto-approve only a HIGH-confidence approval.

ClaimLens never denies a claim on its own.

Every denial and every uncertain case goes to a human adjuster — with the full evidence and the argument already made.

Who it is for

Built for the people the capacity gap actually falls on

Claims adjusters and SIU teams

Primary user. Drowning in volume, with three quarters of flagged claims never fully investigated. Every case arrives pre-argued rather than blank.

Policyholders

Secondary beneficiary. Clear-cut legitimate claims resolve without waiting in a queue behind manual review.

A deliberate product constraint

ClaimLens only ever auto-approves, and only when it can defend every point raised against the claim. Denial is always a human decision — not a limitation, a design choice for a regulated, liability-heavy industry.

What the product does

Four surfaces, one idea: show the uncertainty rather than hide it

- 1 Case queue** Filterable by status, confidence tier and gate outcome. Every row shows whether the cap acted.
- 2 Case view** Evidence in readable cards, the debate as two opposing columns, both Judge orderings side by side with whether they agreed.
- 3 The cap explainer** When confidence is lowered, the page names each unanswered point, what each Judge said about it, and that the claim would otherwise have been auto-approved.
- 4 Portfolio analytics** Accuracy, confidently-wrong counts, auto-resolution, false positives and negatives, fraud value, adjuster hours — reported as raw counts.

Most AI products hide their uncertainty. This one puts it on the page.

Technical architecture

Grounded in published work, not invented for the pitch

Model	openai/gpt-oss-120b on Groq, temperature 0.2. Identical model and settings for the baseline and the full pipeline — the only difference is architecture.
Evidence agent	Fully deterministic, no LLM. Policy timeline, claim history, document consistency, plus live Open-Meteo historical weather calls. Reports neutral facts; never draws conclusions.
Debate layer	A direct application of AI Safety via Debate (Irving, Christiano & Amodei, OpenAI 2018) — verifying a winning argument is easier than verifying a hard claim directly.
Calibration layer	Verbalized confidence cross-checked by an order-swap consistency test, following current work on verbalized-confidence and self-consistency methods.
The cap	Runs in code, not in the prompt. The model cannot argue its way past it.

How we evaluated it

The part that makes every other number on this slide deck checkable

40 synthetic claims

Ground-truth labelled, nine fraud patterns — plus deliberate innocent look-alikes: legitimate claims with new policies, far-from-home incidents and large amounts, each with an innocent explanation. You cannot score well by flagging anything unusual.

Dev / held-out split

13 claims used for every prompt change and debugging decision. 27 never inspected or tuned against.

Pre-registered

The 15-claim evaluation sample was drawn with a fixed seed, stratified, SHA-256 hashed and committed before the first held-out claim ran. So was the HIGH threshold, and the stopping rule.

Methodology log

Every change to inputs or prompts recorded with its justification at the time — including the ones that did not work, and one accidental glimpse of an interim result.

We designed this so we could not fool ourselves. Results reported as raw counts — n=15 does not support percentages.

What we found

ClaimLens did not beat the baseline on accuracy. Here is what it bought, and what it cost.

Correct

Baseline 12 of 15 · ClaimLens 11 of 15

always approve: 10 of 15

Confidently wrong

Baseline 1 of 12 · ClaimLens 0 of 3

ClaimLens before the cap: 2 of 11

ClaimLens auto-approved no fraudulent claims.

Post-hoc comparison, not pre-registered: the same model given the same gate would have auto-approved 11 claims, one of them fraudulent.

The cost of caution

The cap downgraded 8 claims, 6 of which were already correct. Only 2 of 15 auto-resolved.

What didn't work

The order-swap check changed no decisions — 0 of 15. Every tier difference came from the Judge's own stated confidence.

All 15 pre-registered claims completed; nothing was truncated. Every gap here is a single claim — directionally suggestive at most.

What we'd change, and what's next

Known limitations

- The cap is blunt — it caught the two mistakes we wanted and downgraded six correct answers doing it
- The order-swap check changed no decisions; it needs redesigning or removing
- Synthetic benchmark, not real claims data
- 40% fraud rate is deliberate oversampling for signal at $n=40$; real rates are 10-20%
- Small evaluation sample, constrained by free-tier quota
- Document checks are keyword-based and cannot judge whether damage fits a narrative

Roadmap

- Real insurer data and claims-system integration
- Learned calibration — training against an asymmetric reward that punishes confident errors hardest, rather than prompting for it
- Forensics for AI-generated documents and images, where the fraud is heading
- Adjuster feedback loop, so human overrides improve the confidence model
- Audit export, so a regulator can see why any claim was decided as it was

The question for claims AI isn't whether it can decide. It's whether it knows when it shouldn't.